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The enduring benefits of temporary conservation agreements

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ABSTRACT

Short-term conservation agreements (term agreements) are increasingly among the options offered by conservation organizations. Although previous work examined their ability to provide conservation benefits during an agreement, little is known about their possible long-term benefits. Here we investigate the ability of term agreements to provide conservation benefits beyond their conclusion. To achieve this, we examined term agreements offered by Ducks Unlimited Canada for the restoration of annual cropland to tame (non-native) grasslands in the Canadian Prairies. After obtaining land cover data from 2009 to 2023, we compared the occurrence of grassland in sites with term agreements to paired control sites. We found that sites enrolled in term agreements had a significantly higher chance of being majority (>50%) grassland than control sites following the conclusion of the term agreement (53.3% vs. 29.9%). Additionally, most agreement farmlands retain grassland after the end of the agreement for all available years (43.3%; $n = 360$) or some years or quarter-sections (45.2%; $n = 376$). Our results indicate that term agreements provide benefits beyond the period in which habitats are protected under contract. This finding can help conservation organizations prioritize program offerings and optimize the expenditure of limited conservation dollars on permanent options (e.g., easements, fee-simple purchase) and term agreements. Such information can support more strategic allocation of limited conservation resources by identifying when term agreements can complement, or substitute for, permanent conservation options.

1. Introduction

Ecosystem collapse is accelerating worldwide, due to habitat conversion, climate change, and intensification of land use (Díaz et al., 2019; IPBES, 2019). Since many of the world's remaining ecosystems occur on private lands — especially grassland ecosystems — the prevailing pattern of private ownership and management is itself a central constraint on biodiversity conservation (Augustine et al., 2021). In the face of such threats, effective conservation requires creative solutions beyond traditional protected areas that are owned or controlled by conservation agencies or governments (Dinerstein et al., 2017; Maxwell et al., 2020). Conservation organizations and governments often rely on the following three broad approaches to safeguard ecosystems on private land: land acquisition for protected area establishment (fee simple purchase), permanent or long-term covenants that restrict future land use while ownership remains with the landowner (conservation easements), and time-limited stewardship arrangements that incentivize specific management practices for a defined period (term agreements). Since acquisition and easements require considerable financial resources (Wen et al., 2024), they are not always feasible. As such, term agreements may provide cost-effective options to support broader biodiversity goals (Chapman et al., 2023).

Term agreements may also be a valuable mechanism for private land conservation, incentivizing landowners to adopt pro-environmental behaviours for a defined period, without a permanent commitment to limit future land use. They are already used in the Canadian Prairies, a landscape dominated by agricultural production and private land ownership, where conversion of grassland to annual cropland is a leading driver of grassland and biodiversity loss (Caro et al., 2022; Jaureguiberry et al., 2022; Seibold et al., 2019). By encouraging landowners to restore grassland — for example, by converting annual cropland to perennial forage — term agreements can generate meaningful biological benefits (Bortolotti et al., 2022; McMaster et al., 2005) and enhance biodiversity (Augustine et al., 2021; Paterson et al., 2024). Such agreements have delivered concrete outcomes elsewhere: term agreements with rice farmers in California, for instance, created

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temporary wetland habitat for migratory shorebirds (Golet et al., 2018).

Term agreements and related stewardship mechanisms for private-land conservation have been studied across several jurisdictions, including Australia, Europe, and South Africa (Wright et al., 2018). Choice-experiment and stated-preference studies of landholders consistently find that participation depends on more than payment size: contract length and flexibility, administrative and transaction burden, fit with existing operations, and landholders' own stewardship values all shape willingness to enrol (Adams et al., 2014; Greiner, 2015; Moon and Cocklin, 2011; Pittman et al., 2025). A broader literature on the design, delivery, and evaluation of stewardship programs points in a similar direction, linking effectiveness to how schemes are targeted, how simple and flexible their terms are, and how well they align with existing farm operations (Batáry et al., 2015; Cocklin et al., 2007; Dorrough et al., 2008; Franks, 2011; Hajkowicz, 2009; Stoeckl et al., 2015). Yet while the factors shaping participation are well-documented and the short-term biological benefits of increasing grassland area are established, far less is known about the durability of these outcomes once an agreement ends.

Because they are time-limited, term agreements sit outside the criteria for protected areas and other effective area-based conservation measures under the Kunming-Montreal Global Biodiversity Framework, yet may still contribute to several of its targets. A clear understanding of the benefits of term agreements will help organizations strategically support or increase biodiversity in agricultural landscapes and will help policymakers determine their place as a possible support for more permanent measures (Wen et al., 2024). Term agreements as a conservation tool are increasingly studied, particularly through stewardship and incentive program literature (Adams et al., 2014; Kemink et al., 2023). The costs and benefits of term agreements have been documented during specified program periods in government and agency interventions (Golet et al., 2018; Whittington et al., 2019), but it is unclear to what extent conservation benefits persist beyond the program's end in the context of landowner agreements. The difference in program participant commitment is apparent (10 years vs. lifetime or beyond). Whether restored grassland persists once an agreement ends has been examined most closely for the U.S. Conservation Reserve Program (CRP), where a substantial but incomplete share of expired fields remain in grassland after contracts end, with persistence varying across landscapes (Barnes et al., 2020; Sullins et al., 2021; Vandever et al., 2023). To our knowledge, this has not been examined for term agreements delivered by non-governmental organizations or in the Canadian Prairies. Moreover, existing studies rely on landowner self-reports (Barnes et al., 2020) or describe the condition of expired fields without comparison to unenrolled land (Sullins et al., 2021; Vandever et al., 2023). Full knowledge of the benefits of term agreements will contribute to better resource allocation, clear decision-making, and an effective use of limited conservation dollars.

Here we use a case study of term agreements to restore tame (non-native) grasslands to investigate whether term agreements can have a lasting effect beyond their completion. Our research questions are: RQ1: What proportion of quarter-sections maintain grassland area at or above agreement levels after the end of the agreement period? RQ2: Do lands enrolled in term agreements show higher persistence of grassland coverage in the years following agreement expiration compared to control lands? To address these questions, we analyzed land cover data for 10-year "Forage agreements" administered by Ducks Unlimited Canada (DUC) both during and after agreements. We analyzed whether the grassland remained on the landscape post-agreement or was converted back to annual crop (hereafter "reversion"), and whether the likelihood of grassland presence in lands that had previously been enrolled in a term agreement was different from lands that were never enrolled in a term agreement. We expected that a substantial portion of agreement farmland would maintain grassland after the agreement period (RQ1), and that lands previously enrolled in a term agreement would have a higher likelihood of grassland presence than non-agreement lands (RQ2).

2. Methods

2.1. Study system

We analyzed 21 years of term agreements administered by DUC under the "Forage Program" (start dates between 1994 and 2015). The DUC Forage Program provides financial incentive for private landowners to convert annual cropland to tame perennial grassland for a period of 10 years, and sometimes enrolls existing wetlands, native grassland, and trees on the quarter-sections under agreement for protection for that same period of time. The agreements prohibit conversion of the grassland (or other habitat acres enrolled in the agreement) to annual cropland during the term of the agreement, but landowners maintain control over management of the grassland — typically, the grasslands are grazed by cattle or hayed. DUC provided a dataset documenting 885 individual agreements (1679 quarter-sections) covering the Canadian provinces of Alberta, Saskatchewan and Manitoba. Since our primary goal was to examine the retention of grassland following the conclusion of an agreement, it was beyond the scope of this study to perform a thorough examination of the impacts of agreements on grasslands during the agreement period.

2.2. Units of analysis

Two spatial units were considered in this study. The **agreement** is the contractual unit and may comprise one or more **quarter-sections** (805 m × 805 m or 160 acres) of land – the basic unit of land ownership in the Canadian prairies (McKercher and Wolfe, 1986). Agreements are recorded in the source data as one or more project-number records (3457 in total); these are administrative sub-agreements within a single agreement and are not an analytical unit. Of the 885 agreements analyzed, 560 (63.3%) recorded a single quarter-section and 325 (36.7%) recorded more than one. Some agreements may have spanned multiple quarter-sections, but DUC only recorded one, or lumped several smaller agreements into the same quarter-section. Analyses were structured to account for this by making year-over-year comparisons responsive to the composition of a given quarter-section (e.g. using relative targets for reversion; see Section 2.5). We used the quarter-section as the unit of measurement for three reasons. First, enrolment, land cover, and the terms of each agreement are all defined at this scale, so it is the finest resolution at which treatment and outcome are jointly observed. Second, cropping decisions in this system are made at this scale: a landowner may return one quarter-section to annual crop while retaining forage on another. Third, and consequently, aggregating to the agreement before analysis would obscure partial reversion, which we show below to be a common outcome (Section 3.1) and which differs systematically with agreement size (Appendix H).

Reporting units differ between the two components of this study. For the reversion analysis (RQ1), we classified each quarter-section and then summarized at the agreement level, so that agreements with multiple quarter-sections did not disproportionately affect the results. For the post-agreement impact evaluation (RQ2), models were fitted to quarter-section–year observations. Because quarter-sections belonging to the same agreement are not independent, we refitted all three models with an additional random intercept at the agreement level. Between-agreement variance was modest and smaller than between-matched-set variance at every threshold, and the estimates were effectively unchanged, so we report the simpler specification throughout (Section 2.6; Appendix H).

2.3. Experimental design

The experimental design of this study involved two components: a reversion analysis and post-agreement impact evaluation (Fig. 1).

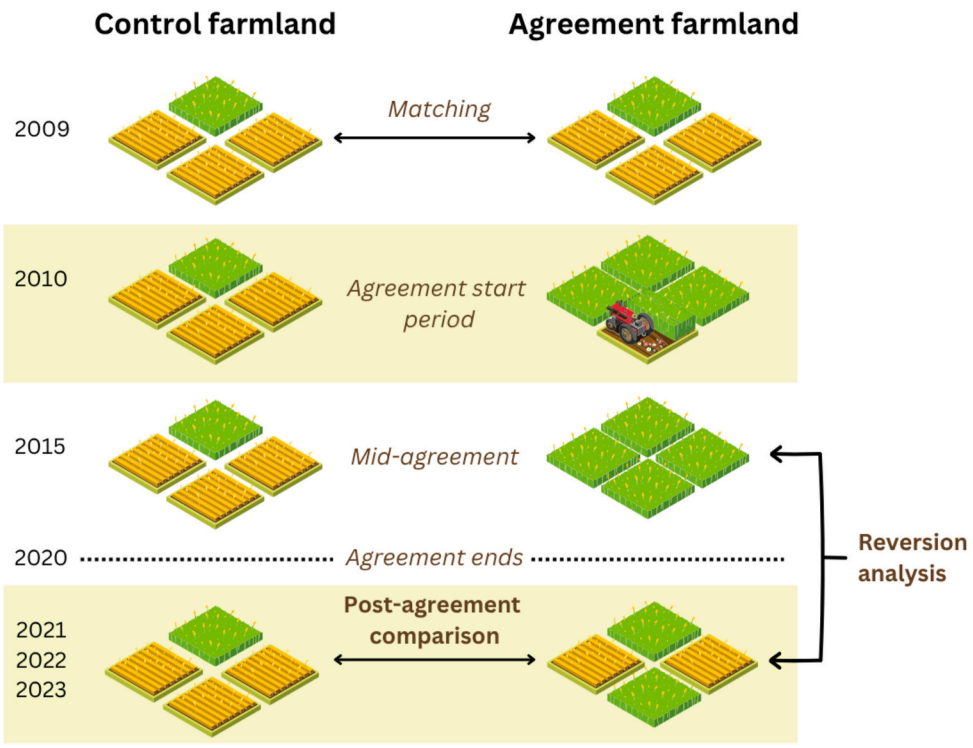


Fig. 1. Schematic of the study design, illustrated for agreements beginning in 2010 and ending in 2020. Rows are years; the left column shows matched control quarter-sections and the right column agreement quarter-sections. Control and agreement sites were matched on 2009 land cover, before any agreement began (top). The reversion analysis (right bracket) compares each agreement quarter-section's post-agreement land cover against its own late-agreement baseline. The post-agreement impact evaluation (bottom) compares agreement quarter-sections with matched controls in each year following expiry. Shaded bands mark the agreement period and the post-agreement period.

Reversion (defined in Section 1) was operationalized as a simultaneous decrease in grassland cover and increase in crop cover relative to conditions during the agreement; the measurement rule is given in Section 2.5. The reversion analysis quantified the extent to which quarter-sections enrolled in the Forage Program were maintained as grassland or converted back to annual crop production following the conclusion of agreements. This provided a direct measure of outcomes associated with the program and allowed us to evaluate the durability of grassland retention on lands enrolled in term agreements. The post-agreement impact evaluation assessed whether observed land cover patterns on enrolled quarter-sections differed from what would have occurred without an agreement. This was necessary because landowners may convert their cropland to grassland without signing up for a program (e.g. to support a cattle herd expansion). To account for the background rate of grassland establishment, we constructed a counterfactual comparison using matched control quarter-sections selected to resemble Forage sites prior to enrollment. Together, these two analyses enabled us to characterize land cover and the extent to which these patterns can be attributed to participation in the Forage Program. All spatial analyses were performed using ESRI ArcGIS (Esri Inc., 2025) and statistical analyses were completed using the R statistical computing environment (version 4.5.0; R Core Team, 2025) on the Nibi supercomputer at the University of Waterloo.

2.4. Data

We used land cover data to characterize agricultural management in the study area. To achieve this, we obtained high resolution (i.e., 56 m for 2009 and 2010, and 30 m for subsequent years) information on the dominant crop type grown in locations across the study area (Agriculture and Agri-Food Canada's Annual Crop Inventory (ACI) (Agriculture and Agri-Food Canada, 2026)). After obtaining these data, we reclassified areas into the following categories: grassland (ACI categories: 110: Grassland; 122: Pasture/forages), natural areas (80: Wetland; 85: Peatland, 200: Forest; 210: Coniferous; 220: Broadleaf; 230: Mixedwood; 50: Shrubland), annual crops, and other (130: Too wet to be seeded; 30: Exposed land/barren, etc.) (see Appendix A for full details).

We prepared geospatial data linking agreement records to land cover classifications for each quarter-section and year. Although records were anonymized by removing the names of participants, geospatial coordinates and information on date of signing and date of agreement expiry were preserved to facilitate subsequent analysis. DUC staff prepared a layer in ArcGIS that contained quarter-sections tagged with the project number for each term agreement. In some cases, agreements spanned multiple quarter-sections. We then used ArcGIS tools to subset the ACI raster files (clip raster) and merge the data with agreements (overlaid quarter-sections and used to calculate portion of grassland). The resulting dataset — consisting of one record per quarter-section per year, each containing the proportion of grassland coverage — was then cleaned and analyzed. Cleaning involved transforming the data by aggregating raster cells and attaching them to specific quarter-sections, which were associated with project numbers. Each quarter-section-year combination was treated as a single observation and duplicates were removed.

2.5. Reversion analysis

To address RQ1 — the proportion of enrolled land that maintains grassland after expiry — we classified the land cover trajectory of each enrolled quarter-section after the conclusion of the agreement relative to its own late-agreement condition. We examined the composition of agreement lands to determine whether grassland remained, or was converted back to annual cropland, after the conclusion of the term

agreement.

For each quarter-section we calculated a late-agreement baseline: the mean proportion of each land cover category — "grassland", "crop", "natural areas" and "other" (Appendix A) — across the final three years of the agreement. For the 13 records (10 agreements) expiring in 2009 or 2010, only one or two of these years fall within the ACI series, and the baseline is the mean of the available years. We used the end of the agreement term rather than its start because seeded perennial forage takes several years to establish and to be reliably classified as grassland, so early-term observations likely understate the grassland actually present. Averaging across three years further reduces the influence of year-to-year classification error and of weather-driven variation in the land cover classification. This baseline therefore represents the established, under-agreement condition against which post-agreement land cover is compared. Proportions were calculated over the full area of each quarter-section, with wetland, forest and shrubland pooled into a separate "natural area" category (Appendix A) rather than excluded. Quarter-sections with extensive wetland or woodland therefore have a lower maximum attainable grassland proportion. This does not affect the reversion classification, which compares each quarter-section to its own late-agreement baseline. It does compress the range available under the ">90% dominant grassland" threshold in wetland-dense quarter-sections, and we therefore interpret the dominant-grassland model as supporting context rather than as a primary basis for inference. The reversion criterion requires a 15% relative decrease in grassland cover together with a 15% relative increase in crop cover, measured against each quarter-section's own late-agreement baseline. For example, a quarter-section whose late-agreement baseline was 60% grassland and 30% crop would be classified as "reverted" in a given post-agreement year if grassland fell more than 15% below that baseline (i.e. $< 51\%$ cover) and crop rose more than 15% above it (i.e. $> 34.5\%$ cover). This is deliberately conservative relative to the Annual Crop Inventory's stated overall accuracy target of 85%. Of 15,451 post-agreement observations, 1328 (8.6%) fell within ±5% (relative) of the criterion in either dimension, indicating that classifications are not concentrated at the decision boundary. Recomputing the classification at 10% and 20% thresholds moved the proportion of records showing no reversion from 52.5% to 46.2% and 56.6% respectively, and the proportion showing full reversion from 12.7% to 14.8% and 11.3% (Appendix I). The direction and relative magnitude of our findings are therefore stable across this range, though the absolute proportions should be read as accurate to within roughly six percentage points.

We evaluated the reversion of any lands subject to agreements that ended in 2022 or earlier (of the 885 agreements; 1679 quarter-sections; exclusions below) to allow for at least one post-agreement year of observations. To evaluate patterns of post-agreement land reversion, we summarized changes in land cover at the quarter-section level and quantified reversion events. We then calculated the proportion of quarter-sections that experienced reversion in each post-agreement year and summarized these outcomes.

Of 3457 project records, 112 records (3.2%) (53 agreements) could not be classified: 94 had no post-expiry observation year within the ACI series, 12 lacked a late-agreement baseline, and 6 lacked an expiration date. These records were excluded from the reversion analysis. Agreements were classified as showing no reversion where no constituent quarter-section met the reversion criterion in any observed post-agreement year; full reversion where every constituent quarter-section met it in every observed post-agreement year; and partial reversion otherwise — including agreements in which some quarter-sections reverted and others did not, and agreements in which a quarter-section met the criterion in some years but returned to grassland in others.

2.6. Post-agreement impact evaluation

To address RQ2 — whether enrolled land retains grassland at a higher rate than comparable unenrolled land — we constructed a matched counterfactual and estimated the difference in grassland probability in each year before, during and after the agreement period. We performed a post-agreement impact evaluation analysis based on the counterfactual approach to examine the effect of establishing agreements on land management (reviewed in Ribas et al., 2021). This was particularly important for our study system because landowners may establish forage on their land independent of the DUC Forage Program (e.g., for haying or grazing of cattle). For the purposes of this model, we scoped the sample to focus on agreements that commenced in 2010 and ended in 2020 ($n = 163$), allowing for 3 years of post-agreement analysis using land cover data. For each quarter-section enrolled in the Forage Program (hereafter, "agreement farmland"), we identified a nearby counterfactual quarter-section that was not enrolled in an agreement (hereafter, "control sites"). By analyzing differences between the agreement sites and the control sites, we inferred the impact of establishing agreements on land management.

We identified control sites using a matching method for causal inference (reviewed by Stuart, 2010). To achieve this, we first restricted candidate control sites to the Prairie Habitat Joint Venture target areas spanning the provinces of Alberta, Saskatchewan and Manitoba (Prairie Habitat Joint Venture, 2026), because the majority of agreement farmland falls within them. These target areas encompass productive grassland and shrub-steppe habitats supporting significant waterfowl and grassland bird populations (Fig. 2). We then excluded candidate sites that were (i) an agreement site, (ii) currently enrolled in any DUC program or owned by DUC, or (iii) within 5 km of an agreement site to decrease the likelihood that control and agreement farmland were owned by the same landowner. A distance of 5 km was chosen based on typical landholding patterns in the Canadian Prairies, though we acknowledge the buffer may not prevent spillover for some distant non-contiguous landholdings. We judged this an acceptable trade-off: a larger buffer would have improved separation at the cost of excluding areas of highest agreement density, biasing the control pool toward landscapes systematically unlike those where agreements occur. Next, we used the fast generalized full matching method to match each agreement site with a set of control sites and obtained a numerical weight for each set of sites (with the *MatchIt* R package, Ho et al., 2011). In particular, this method involved computing propensity scores based on a generalized linear model fitted with 13 covariates recorded during 2009 including crops (wheat, canola), natural features (water, broadleaf forest, coniferous forest, mixedwood forest, wetland, shrubland) and grassland. We transformed the continuous grassland into binary threshold variables indicating whether the site had any grassland, more than 50% grassland or more than 90% grassland and included those as matching covariates. Thus, after completing the matching method, we obtained a set of control sites for each agreement site, along with weights describing their similarity in land cover before the establishment of agreements on the agreement farmland. We acknowledge that this transformation discards information about the degree of grassland coverage, potentially reducing matching quality (Stuart, 2010). We nonetheless retained the threshold specification because the outcomes of interest are themselves defined at these thresholds, so matching on the same thresholds ensures treated and control sites are comparable with respect to the quantity being estimated. The resulting matched sample provides estimates of the Average Treatment Effect on the Treated (ATT), which is appropriate for evaluating the effect of the Forage Program for landowners who elected to participate. We acknowledge that a more comprehensive counterfactual would require additional confounding variables (e.g., landowner demographics, economic factors, social influences) that we were unable to measure.

We fitted generalized linear mixed effects models to conduct the impact evaluation analysis (with the *lme4* R package, Bates et al., 2015). In particular, we fitted three models with response variables indicating if grassland coverage at a given site recorded during a given year exceeds 0% (hereafter, "any grassland"), 50% (hereafter, "half grassland"), or

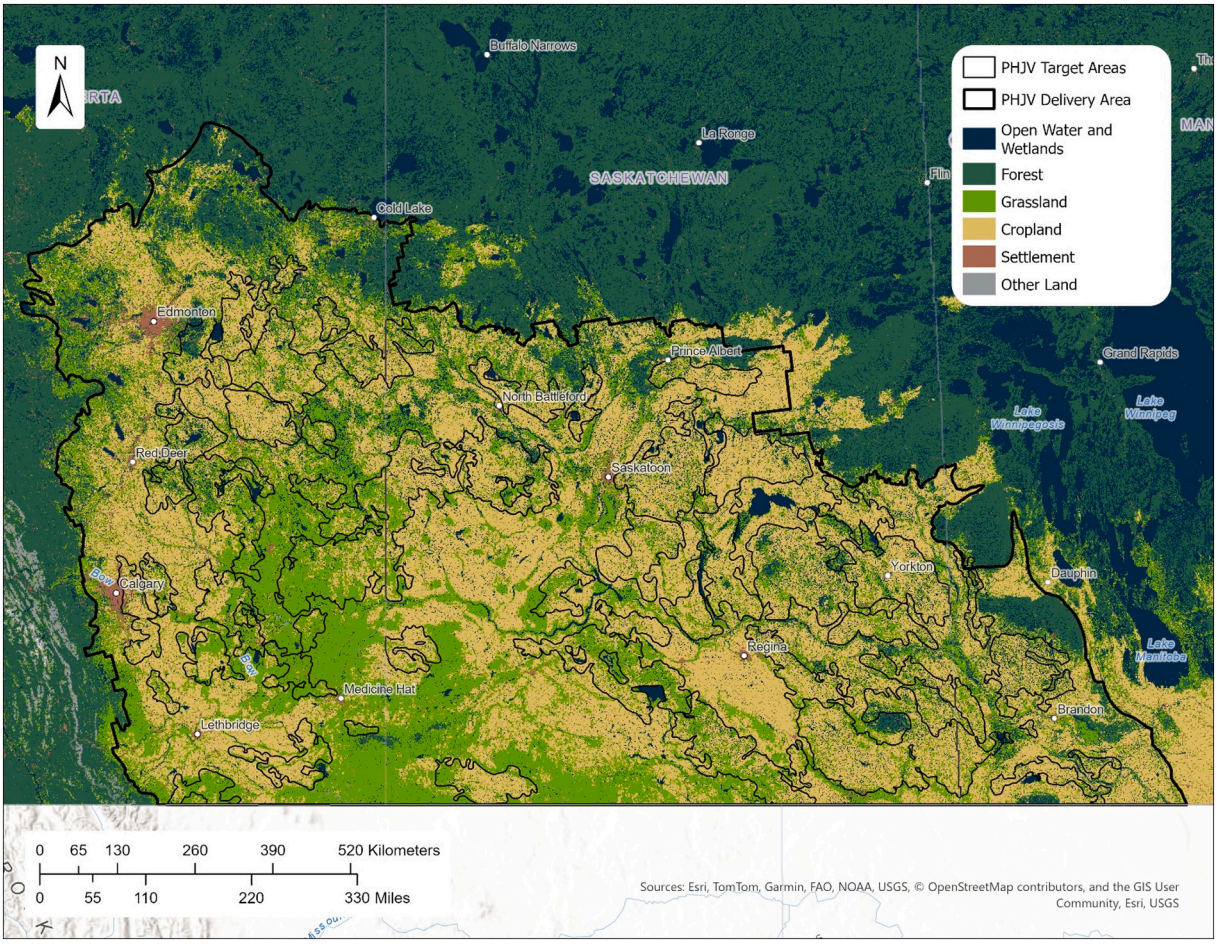


Fig. 2. Prairie Habitat Joint Venture (PHJV) target areas and delivery area across the Canadian provinces of Alberta, Saskatchewan and Manitoba, shown over generalized land cover. Candidate control quarter-sections were drawn from within the PHJV target areas. Individual quarter-sections (805 m × 805 m) are not resolvable at this extent and are not shown; quarter-sections within 5 km of an agreement site were excluded from the control pool during analysis (Section 2.6).

90% (hereafter, "dominant grassland"). These models were fitted with categorical predictor variables (fixed effects) indicating the year of observation (as a proxy for the temporal phase of the agreement, rather than the grassland coverage used as a response variable), if the site was a control or agreement, and the interaction between them. After fitting these models, we performed likelihood ratio tests as implemented in a sequential term deletion routine to assess the significance of these predictor variables. All models were fitted with a binomial error structure, a random intercept term for matched set to ensure comparisons were made between matched treatment and control sites, and numerical weights from the matching procedure. Because an agreement may enrol more than one quarter-section — the 163 agreements in the impact-evaluation sample comprise 273 quarter-sections (mean 1.67, median 1, maximum 10; 68.1% of agreements enrolled a single quarter-section; Appendix H) — quarter-sections under the same agreement are not statistically independent. We therefore refitted all three models with an additional random intercept at the agreement level, assigning control quarter-sections to the agreement of the treated site they were matched to, so that matched sets are nested within agreements. Between-agreement variance was modest and smaller than between-matched-set variance at every threshold (0.70 versus 1.68 for "Any (>0%)", 0.32 versus 1.98 for "Half (>50%)" and 0.17 versus 2.24 for "Dominant (>90%)"), and the estimates were effectively unchanged: across all 18 within-year contrasts the largest difference was 0.005 on the log-odds scale, standard errors differed by at most 0.1%, and no contrast changed its significance status (Appendix H). We therefore report the simpler specification throughout.

The primary models were fitted with the Bound Optimization By Quadratic Approximation (BOBYQA)and bounded Nelder-Mead algorithms respectively (via the *minqa* and *dfoptim R* packages; Bates et al., 2024; Varadhan et al., 2023), and allowing for a maximum number of 200,000, 1×10^{10}, and 2×10^{10} iterations, respectively. The agreement-level refits reported above were fitted with BOBYQA throughout and converged without warnings. We then generated within-population 95% confidence intervals for the model predictions (Roy et al., 2024) using estimated marginal means (with the *emmeans R* package; Lenth et al., 2025) with bias adjustment using each model's random-effect standard deviation extracted from the variance components (with the *lme4* R package, Bates et al., 2015). We then compared results within years to determine if there was a statistically significant difference between treatment and control predictions in each year (with the *emmeans R* package; Lenth et al., 2025). Before interpreting model outputs, we assessed matching performance using standardized mean differences, treated/control covariate means, propensity-score overlap, empirical cumulative distribution summaries, and matching-weight distributions (Appendix F).

3. Results

The final dataset contains forage agreements that cover 84,859 ha across all three prairie provinces. The agreement start dates range from July 1994 to March 2015, with an average of 95.9 ha per agreement. Overall, 68,865 ha of grassland were planted and 15,994 ha of habitat (wetlands, native grassland, trees) were temporarily protected as part of

the agreements. The mean proportion of grassland was consistently higher in the agreement farmland group compared with the control group throughout the study period (Appendix C). Classifiable agreements were observed for a mean of 4.94 post-agreement years (median 5; range 1–14; $n = 832$ agreements). Post-agreement follow-up in the impact-evaluation subsample is shorter and fixed by design: agreements in that sample ended in 2020 and were observed for three subsequent years (2021−2023).

3.1. Post-agreement grassland retention (RQ1)

We analyzed a total of 832 agreements. Most retained grassland at a level comparable to the agreement period for at least part of the observed post-agreement period: 360 agreements (out of 832 classifiable agreements; 43.3%) showed no reversion in any observed year, and a further 376 (45.2%) showed partial reversion, reverting on some quarter-sections or in some years but not others. A total of 96 agreements (11.5%) reverted fully to annual crop across all constituent quarter-sections and all observed post-agreement years. A total of 53 agreements could not be classified. Reversion outcomes differed by agreement size. Of the 517 agreements comprising a single quarter-section, 249 (48.2%) showed no reversion, 189 (36.6%) partial and 79 (15.3%) full. Of the 315 agreements comprising more than one quarter-section, 111 (35.2%) showed no reversion, 187 (59.4%) partial and 17 (5.4%) full (Appendix H). Larger agreements were therefore less likely to avoid reversion entirely, but also less likely to revert fully — full reversion requires every constituent quarter-section to revert in every observed year, which is a higher bar for a larger agreement. Both patterns are observable only because land cover was classified at the quarter-section scale (Section 2.2).

3.2. Grassland persistence relative to matched controls (RQ2)

We found that enrolled land retains grassland at a higher rate than comparable unenrolled land in every post-agreement year and at every grassland threshold examined. The binomial GLMM showed statistically significant differences within years between agreement and non-agreement farmland both within and outside the term period (Fig. 3). We first assessed whether the matching procedure produced comparable treatment and control groups. Before matching, five covariates showed standardized mean differences (SMDs) above 0.2. After matching, all SMDs fell below the conventional 0.1 threshold, with a maximum of 0.06, including the dominant-grassland indicator (−0.02; Appendix F). Differences between agreement and control farmland were not statistically significant in 2009, before the start of any agreement, at any of the three grassland thresholds (Any: $p = 0.16$; Half: $p = 0.17$; Dominant: $p = 0.59$), as expected from the matching procedure. We nonetheless interpret the "Dominant (>90%)" model as supporting context rather than as a primary basis for inference, because quarter-sections containing substantial wetland or woodland cannot reach the >90% grassland threshold regardless of management (Section 2.5), which restricts the population to which that model applies. The matching procedure assigns highly variable weights to control quarter-sections. Residual diagnostics and weight-sensitivity checks confirmed that the results are not driven by a small number of high-weight observations (Appendix G).

We are confident that our models are suitable for inference. This is because all three models converged, and none of them showed singular fits or boundary estimates (Table 1). The matched-set random-effect variances were substantial and near-identical across different thresholds, indicating the grouping structure is well identified (Table 1). Furthermore, because the random-effect variance is well away from zero in every model (all >2.3), this provides evidence that the matched sets are informative as a grouping factor. Finally, Pearson-residual-versus-fitted plots (Supplementary Fig. G1–6) showed no systematic departure from zero across the fitted range in any model, indicating the mean structure and link function are appropriate.

Following the start of agreements, models predicted a substantially higher proportion of agreement farmland would meet each grassland threshold compared with controls. The model predicted a higher proportion of agreement farmland would be any, dominant or half grassland than controls mid-agreement.

Increases in grassland coverage associated with term agreements persisted beyond the conclusion of the agreement period. There continued to be a statistically significant difference between the predicted control and agreement proportion of quarter-sections in grassland post-agreement. For instance, the proportion of agreement sites predicted to be more than half grassland remained close to twice that of control sites in the post-agreement period (e.g., 45.1% vs. 24.1% in 2022). All models showed a statistically significant difference between treatments and controls for a given year (Appendix E). The magnitude of the agreement effect varied across thresholds: in absolute terms, the

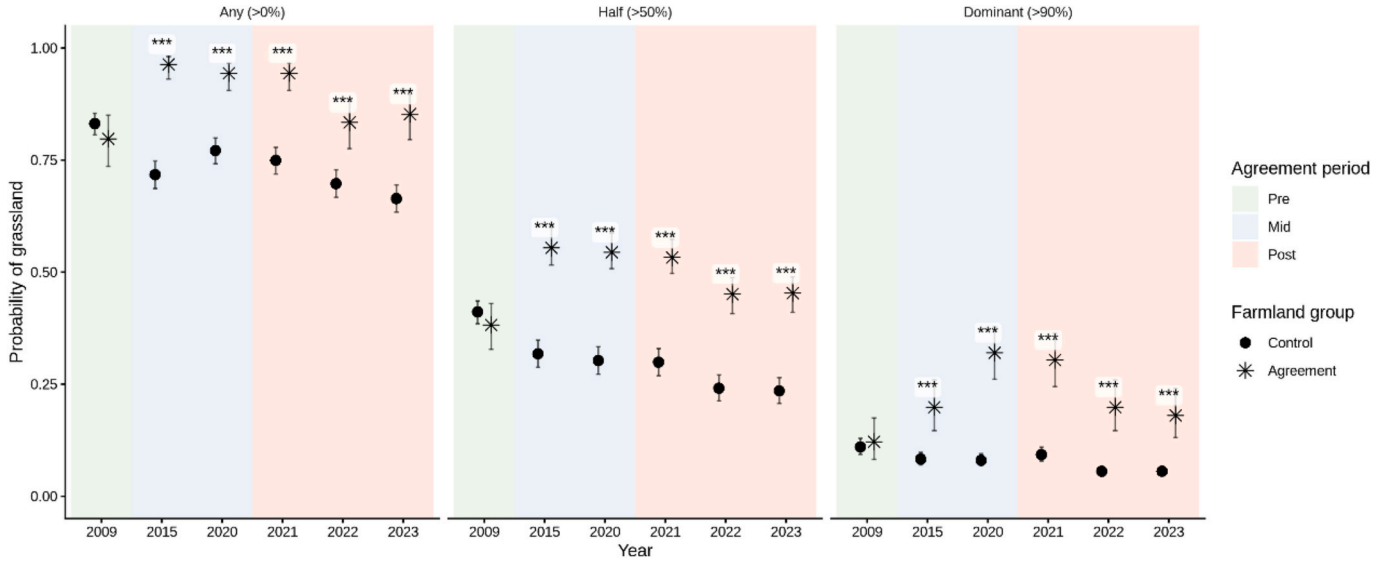


Fig. 3. Model predictions for agreement and control quarter-sections. Models are based on agreements in place from 2010 to 2020. Asterisks indicate a statistically significant difference between treatment and control sites in a given year (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$). Error bars indicate 95% confidence intervals. See Appendix E for raw values.

Table 1
The table shows extended model statistics for each of the three models. Re_variance = random effect variance; re_sd = random effect standard deviation; obs = n of observations; pearson_csq = Pearson chi-squared; df_approx = approximate degrees of freedom; dispersion_ratio = approximate dispersion ratio. All models converged successfully. Models include a single random intercept, for matched set.

Model	re_variance	re_sd	obs	pearson_csq	df_approx	dispersion_ratio
Any (>0%)	2.3676	1.5387	656,870	517,077.54	656,858	0.7872
Half (>50%)	2.3253	1.5249	656,870	561,693.37	656,858	0.8551
Dominant (>90%)	2.4163	1.5544	656,870	533,575.33	656,858	0.8123

"Half (>50%)" model showed the greatest divergence between agreement and control sites, while the "Any (>0%)" model showed the smallest difference, suggesting that agreements may be particularly effective at maintaining moderate levels of grassland coverage (Fig. 3). There was a noticeable decrease in predictions for agreement quarter-sections in 2022 (i.e., two years after the conclusion of the agreements), though the difference from controls remained significant across all models.

4. Discussion

We evaluated the proportion of enrolled lands that maintain grassland at or above agreement levels after the agreement ends (RQ1) and whether enrolled lands show higher grassland persistence post-expiry compared to control lands (RQ2). Specifically, we investigated whether lands enrolled in the DUC Forage Program retained grassland after agreements ended, and whether post-agreement grassland persistence exceeded levels observed on matched control lands. We found evidence for both. Most enrolled agreements retained grassland for at least part of the post-agreement period, and agreement quarter-sections remained substantially more likely than matched controls to contain grassland after the agreement concluded. These findings suggest that term agreements can produce conservation outcomes that outlast the contract period, although the durability and mechanism of this effect remain uncertain. If these outcomes prove durable, term agreements would deliver conservation value well beyond their contract period, reinforcing their promise as a cost-effective means of advancing biodiversity goals on private land where acquisition and easements are not feasible.

Previous work has shown that private land conservation programs can influence landholder participation, program uptake, and conservation outcomes during active agreements (Adams et al., 2014; Kemink et al., 2023; Stoeckl et al., 2015). Less is known about what happens after temporary agreements end. Recent work also indicates that the persistence of conservation and climate-smart practices after adoption is variable and often incomplete across a range of U.S. programs (Ferraro et al., 2025), consistent with the partial persistence we observe here. Our reversion analysis addresses this gap by tracking whether enrolled lands return to annual cropland after expiry, while our matched comparison evaluates whether post-agreement grassland persistence was greater than would be expected on comparable non-enrolled lands.

Our model suggests that agreements increase the likelihood of grassland persistence beyond that of comparable agricultural land. This result echoes and extends a growing body of work on the long-term fate of grasslands established under temporary incentive programs. Sullins et al. (2021) estimated that 58% of expired fields remained grassland a decade after U.S. CRP contracts ended. Vandever et al. (2023) found that 41% retained at least half their area in conservation cover three or more years post-contract, and Barnes et al. (2020) reported comparable rates of landowner-reported persistence. Our estimate that enrolled quarter-sections have a 53.3% probability (upr: 57.3%; lwr: 49.7%) of remaining majority-grassland in the year after expiry falls within this range, suggesting that the partial durability documented for U.S. set-aside programs also applies to NGO-delivered term agreements in the Canadian Prairies. This estimate is the model-predicted probability for the first year after expiry; it is a year-specific prediction and does not require that the site remain in grassland in subsequent years. Our design differs from this prior work in a key respect: whereas these studies describe the condition of expired fields on their own, our matched comparison estimates persistence relative to comparable unenrolled land, isolating the signal attributable to the agreement. At the same time, this literature cautions that persistence is neither complete nor permanent: Barnes et al. (2020) and Sullins et al. (2021) found reversion more likely as time since expiration increases, and reported persistence proportions ranging from roughly one-fifth to over four-fifths of fields across regions — reinforcing that our positive post-agreement signal should not be read as permanence.

Among records classified as reverting and observed for at least two post-agreement years ($n = 1429$), 26.5% reverted in no more than two years and were classified as grassland in every remaining year, rather than remaining in annual crop. Grassland stands may need to be periodically "rejuvenated" in order to continue producing forage of sufficient quality and quantity to maintain cattle — a process that may involve "breaking" the grassland (i.e., plowing it) and re-seeding the area, often to annual crop for 1–3 years before returning it to grassland (Omokanye et al., 2018). Slightly over a quarter of reverting records follow this pattern, so rejuvenation is a plausible explanation for part but not most of the movement we observed between crop and grassland. We cannot distinguish it from other short-term rotations, or from reversions followed by re-establishment for unrelated reasons, and we did not observe management directly (Appendix J). During such a time, our analysis would register a reversion year or years for the land, followed by grassland years. The reversion results also suggest that post-agreement outcomes are not binary. Some lands appear to remain continuously in grassland, some return fully to annual crop production, and others move between crop and grassland during the post-agreement period. This pattern is important because temporary conversion to annual crop may sometimes reflect pasture rejuvenation rather than permanent abandonment of grassland management. Distinguishing true reversion from rotational or rejuvenation practices is therefore an important next step for evaluating the durability of grassland agreements in working agricultural landscapes. Overall our findings are consistent with evidence that participation in conservation programs can shift landowner attitudes and management norms over time (Reimer, 2015; Sorice et al., 2013), though the mechanisms underlying post-agreement outcomes warrant further investigation.

Our findings align with a broader international literature showing that the success of private-land conservation depends on landholder motivations, program design, opportunity costs, and institutional fit. Studies from Australia have shown that willingness to participate in conservation contracts is shaped by landowner attitudes, perceived restrictions, payment levels, and compatibility with farm operations (Cocklin et al., 2007; Greiner, 2015; Moon and Cocklin, 2011). Program design studies similarly emphasize that conservation incentives must account for trade-offs between ecological benefits and farm-scale economics (Dorrough et al., 2008; Stoeckl et al., 2015). Recent impact evaluations of private-land conservation have also highlighted the importance of explicit counterfactual design when estimating program effects (Kemink et al., 2023). Our results contribute to this literature by showing that temporary agreements may leave a detectable post-contract land-cover signal in the Prairies. However, because landowner decisions are shaped by economic conditions, social norms, and

management goals that were not observed in this study, these results should be interpreted as evidence of durable land-cover association rather than proof of durable behaviour change.

Term agreements may advance several goals of the Kunming-Montreal Global Biodiversity Framework (Global Biodiversity Framework, 2022). In particular, Target 1 calls for integrated spatial planning that balances conservation and sustainable use. Term agreements can enable near-term integrated spatial planning while larger planning processes are being developed. In connection with Target 4, which calls for urgent management actions to halt human-induced extinction, term agreements present a significant opportunity to implement active management interventions that support biodiversity. Finally, in relation to Target 10, which calls for the sustainable management of agricultural areas to increase biodiversity while preserving food security, term agreements present a pathway for incentivizing landowners to convert annual crops to grassland, providing benefits for grassland biodiversity while keeping the land in agricultural production. Because they are time-limited, term agreements do not meet the criteria for other effective area-based conservation measures (OECMs) under frameworks such as the Kunming-Montreal Global Biodiversity Framework, and so do not contribute toward area-based conservation targets (Watson et al., 2023). They nonetheless offer landholders a flexible mechanism for adopting pro-environmental land management.

Conversion of annual cropland to grassland is a part of farming practice in the Canadian prairies, particularly among cattle farmers and mixed crop operations. In some cases, grassland may be a part of a crop rotation plan (Entz et al., 1995). The choice of whether to establish more grassland on a working farm can have agronomic, economic, psychological, social and other influences. These decisions reflect more than economics alone: alongside the opportunity cost of foregone cropping, landholders weigh their own attitudes, values, and stewardship motivations, and these vary considerably from one operation to the next (Greiner, 2015; Moon and Cocklin, 2011). This heterogeneity means that uptake is seldom driven by payment levels alone, and instead depends on how well an agreement's terms and delivery align with a given landholder's goals and circumstances. Conservation agreements like the DUC Forage Program must balance restrictiveness, incentives and trust to achieve a fit with working landscapes (Pittman et al., 2025).

Conservation organizations increasingly use spatial prioritization tools to guide large-scale investment, weighing the monetary value of land, species distributions, and other features to identify where limited dollars deliver the greatest benefit (Margules and Pressey, 2000; Naidoo et al., 2006). These tools — in use at major organizations including Ducks Unlimited Canada (Devries et al., 2023) and the Nature Conservancy of Canada (Hanson et al., 2024) — perform best when the costs and benefits of the available conservation options are well-characterized. Yet prioritizations are typically built on assumptions and data for permanent protection, which can yield very different priority areas than if temporary options are also considered. Where cost is high and feasibility is low, areas with relatively high biodiversity may be excluded from priorities (e.g., Proctor et al., 2022). By demonstrating that term agreements can protect grassland beyond the contract period, our findings give organizations a firmer basis for valuing these temporary options. This matters most in working landscapes such as the Canadian Prairies, where term agreements may extend the feasibility of conservation delivery into areas that permanent protection cannot reach. Accounting for the durability of term agreement benefits alongside permanent measures would let organizations allocate limited resources more strategically.

Importantly, our analysis does not account for unmeasured confounding factors that may influence both program enrollment and post-agreement land use decisions. Economic factors (e.g., crop prices, commodity markets), political factors (e.g., government policy shifts, subsidy programs), and social factors (e.g., landowner demographics, network effects, neighborhood norms) may all contribute to landowner behaviour independent of the agreement itself. While matching addresses observable confounders relating to land cover history, it cannot control for unmeasured factors that differ between participating and non-participating landowners. Future work could expand the counterfactual analysis to include measures of landowner characteristics, economic conditions, and social influences where such data are available. The decline in predicted grassland on agreement quarter-sections in 2022 (Fig. 3) coincides with two Prairie-wide shocks that we did not measure but that plausibly bear on it. The severe drought of 2021 produced widespread forage and feed shortages across the three provinces and prompted early cattle sales and herd reductions (Government of Canada, 2022). Grain and oilseed prices rose sharply into the 2022 crop year — farm cash receipts for wheat and canola were driven by price increases of 42.0% and 38.9% respectively, despite lower marketings (Government of Canada, 2023) — raising the opportunity cost of holding land in forage at the point when 2022 seeding decisions were being made. Both would be expected to increase conversion to annual crop on the timescale we observe (Fig. 3). We emphasize that we did not test these mechanisms: our data cannot distinguish them from one another or from ordinary pasture rejuvenation, and we raise them as hypotheses for future work rather than as explanations.

Given the window of availability for ACI data (15 years; 2009 to 2023) and the length of the agreement (10 years), it was not possible to systematically analyze landscape composition for long periods of time before and after the agreements. Thus, we were unable to assess how long the land enrolled in the Forage Program was annual cropland before enrollment and may have been unable to distinguish reversion from rejuvenation post-agreement (i.e., in more recent agreements where there were only 1–2 years of ACI data post-agreement). Similarly, our matching analysis could not account for pre-2009 land cover trajectories. The ACI is compiled annually and is subject to error. Although the ACI has an overall accuracy target of 85% (Agriculture and Agri-Food Canada, 2026), accuracy of individual land cover classes may be lower. Despite these limitations, the ACI remains the only consistent, annually updated, geographically complete land cover product that covers the full study period and spatial extent, and the alternative — landowner self-report — carries its own recall and social-desirability biases and cannot support a matched counterfactual design. Our reversion criterion was set well above the expected classification error rate for this reason. Additionally, the nature of grassland is such that it takes time to establish on the landscape. Therefore, measurements during the early years of grassland establishment may be unreliable. Similarly, a landowner may have converted land to crop at the conclusion of the agreement, but that land may have been grassland for part of a year, leading the ACI to misclassify it. While we accounted for error in our reversion analysis, the analysis does not distinguish landowners who partially revert their quarter-section from those who fully revert. DHARMa's dispersion and outlier tests (Appendix G) are statistically significant given the very large sample size but the effect sizes (ratios $\approx$ 1, KS D < 0.01) indicate no meaningful lack of fit.

Several features of our study point to productive directions for future research. Because we could not assess whether rotation away from grassland and back again was already part of a landowner's practice before enrollment, longitudinal designs that capture pre-agreement management histories would distinguish genuine post-agreement persistence from pre-existing rotational patterns. Our focus on tame (non-native) grassland opens a comparative avenue: while restoring native grassland is more valuable for biodiversity, the grazing systems the Forage Program promotes are themselves beneficial relative to annual crops (Fraser et al., 2014; Paterson et al., 2024), and future work could weigh the relative feasibility and benefits of conversion to tame grass against restoration to native prairie as part of an integrated conservation strategy. Most importantly, the exact duration of the benefit provided by term agreements, and the factors that determine whether landowners maintain grassland, remain open questions. Durability of behaviour change is a long-standing challenge in conservation research

(De Young, 1993). There would be clear value in long-term monitoring to establish how persistent these outcomes are, paired with investigation into why landowners revert grassland. Taken together, this would sharpen the case for where and when term agreements are the right conservation instrument.

We found that measures implemented by temporary grassland restoration agreements tend to stay in place even after the conclusion of the agreements, and the likelihood of grassland persisting remains significantly higher post-agreement as compared with control sites. This means that term agreements may have long-term benefits, which is an important characteristic when prioritizing conservation instruments on the landscape. As ecosystem loss accelerates and much of what remains lies on private land, creative approaches beyond conventional protected areas are essential. Our findings suggest that term agreements, by delivering grassland benefits that can persist beyond the contract period, deserve a place in the conservation toolkit alongside permanent protection and extension programs as one means of slowing biodiversity loss in working landscapes.

CRediT authorship contribution statement

Tim Alamenciak: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Lauren Bortolotti:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Data curation, Conceptualization. **Jeffrey O. Hanson:** Writing – review & editing, Software, Methodology, Formal analysis. **Joseph R. Bennett:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the lead author used Anthropic Claude Opus 4.5 in order to refactor code for use in a high-performance computing environment. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biocon.2026.112177.

Data availability

All code and anonymized data will be uploaded to a Zenodo repository and made available with this article.

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